ADS401 · PRESENTATION 2

Supplementary Materials

Group	**Group C**
Project title	Smart Partner: How XJTLU Year 1 Students Actually Use an AI Learning Agent
Contents	Research questions · Sampling · Study design · Instruments · Information sheet · Consent form · Analysis plan · Reflection · Ethics

Figures marked with a dashed line are placeholders. They are intentionally left blank until the survey has been administered and the results analysed.

1. Research Questions

Q1 **Breadth and depth of use.** Among XJTLU Year 1 students, what is the breadth of Smart Partner use (has a student ever used it?) and its depth (sessions per week, minutes per session)?

Q2 **Differences by EAP level.** Do the four EAP levels — A (foundation), B and C (core), High Level (advanced) — differ significantly in adoption rate and weekly use frequency?

Q3 **Use and confidence.** Is students' use frequency correlated with their self-rated EAP confidence?

Q4 **Motivation.** Do students use Smart Partner mainly as an exam-sprint tool, or to build long-term study habits?

2. Target Group & Sampling Method

2.1 Target group

Respondents are Year 1 students at Xi'an Jiaotong-Liverpool University who are currently enrolled on an EAP module. The sample spans all four EAP levels — **A** (foundation stream), **B** and **C** (core streams, where most students are placed) and **High Level** (advanced stream) — and covers arts, science and engineering programmes, so that the picture reflects the whole Year 1 cohort rather than a single school.

2.2 Sampling method

We used **convenience sampling**, a non-probability method: the questionnaire link was distributed through EAP class group chats and year-group chats. This was chosen because the Year 1 cohort is large and no complete sampling frame was available to us. The recognised cost is that cell sizes are uneven across levels, which is why non-parametric checks are reported alongside the parametric tests and the limitation is stated openly in the conclusions.

Stage	Detail
Population	XJTLU Year 1 students currently taking EAP
Method	Convenience sampling (non-probability)
Distributed	----------
Returned	----------
Valid after cleaning	----------
Response rate	---------- %

Stratification	Reported separately for A, B, C and High Level

3. Study Design

This is a **cross-sectional survey** with a **non-experimental design**: usage and attitude data are collected at a single time point, and no variable is manipulated. We measure differences and associations as they occur naturally, so every finding is correlational rather than causal.

3.1 Experiment plan

Stage	Activity	Output
1	Background review — Smart Partner's positioning and the EAP streaming system	Four research questions fixed
2	Instrument design — structured questionnaire plus semi-structured interview guide	Draft questionnaire, interview script
3	Pilot test with approximately 12 Year 1 students	Ambiguous wording revised; completion time recorded
4	Fieldwork — link distributed via class group chats	Raw responses
5	Data cleaning and coding — incomplete and contradictory responses discarded	Analysis-ready dataset
6	Analysis — descriptive statistics, then inferential tests	Test statistics and effect sizes
7	Reporting — visualisation, presentation and reflection	Slides and this document

3.2 Data types collected

Measurement level	Example variables	Analysis used
Nominal	EAP level, ever used, faculty / school	Counts, percentages, chi-square

Ordinal	Use-frequency band, self-rated confidence (1–10), Likert satisfaction	Median, Wilcoxon signed-rank
Scale	Sessions per week, minutes per session	Mean, SD, one-way ANOVA, Pearson *r*
Textual	Open-ended answers and interview transcripts	Thematic coding

3.3 Why these evaluation methods

Each test was matched to the shape of the data on three criteria: **(i) the measurement level of the variables** — chi-square for categorical pairs, ANOVA or correlation for scale variables; **(ii) the design** — a one-way ANOVA for four independent conditions, a paired test for any pre/post comparison; and **(iii) whether the parametric assumptions hold**, checked with Shapiro–Wilk for normality and Levene for homogeneity of variance. Where an assumption failed, the corresponding non-parametric test was used instead. Because cell sizes are uneven across the four levels, effect sizes (Cramér's V, η^2) are reported alongside *p* values. All tests are two-tailed at $\alpha = 0.05$.

4. Instrument — Questionnaire Script

Structured online questionnaire, approximately 6–8 minutes to complete. Closed items precede the single open item.

Q1 Which EAP level are you in?

A / B / C / High Level

Q2 Have you used Smart Partner?

Yes / No — if "No", skip to Question 8

Q3 How many times a week do you use it on average?

Numeric entry

Q4 How long is a typical session?

<10 / 10–20 / 20–30 / 30–45 / >45 minutes

Q5 Which features do you use most?

Multi-select: grammar practice / writing feedback / vocabulary consolidation / mock EAP tests / academic English Q&A / exam revision planning

Q6 What is your main purpose in using it?

Multi-select (6 items), including "pass the EAP exam" and "build independent study habits"

Q7 How confident are you about passing the EAP exam?

1 (not at all confident) – 10 (extremely confident)

Q8 What is the single biggest improvement Smart Partner needs?

Open-ended, optional

5. Instrument — Interview Guide

Semi-structured interview, approximately 15 minutes, conducted with participants who opted in. Questions are prompts rather than a fixed script; interviewers follow up on whatever the participant raises.

Q1 In what context did you first come across Smart Partner?

Q2 Describe one time it was especially helpful — or especially useless.

Q3 Would you recommend it to a classmate? Why or why not?

Q4 Has it changed how you prepare for EAP? In what way, specifically?

Q5 If one feature could be added, what would you want it to be?

6. Participant Information Sheet

Information sheet

- **Purpose** — to understand how XJTLU Year 1 students actually use and perceive Smart Partner, an AI learning agent designed to support EAP study.
- **What is involved** — an online questionnaire taking approximately 6–8 minutes. Participants may optionally leave contact details for a follow-up interview of about 15 minutes.
- **Voluntary participation** — participation is entirely voluntary. You may withdraw at any time, without giving a reason, and with no effect on any course grade or on your relationship with the university.
- **Anonymity and confidentiality** — no names, student IDs or contact details are attached to responses. Data are used only for this coursework and are reported in aggregate form only.
- **Data storage** — responses are stored encrypted on the researcher's own device and deleted once the module has ended.
- **Contact** — questions about the study may be sent to _____.

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7. Consent Form

Consent form

Please tick each statement before proceeding to the questionnaire.

- ☐ I have read and understood the participant information sheet.
- ☐ I understand that participation is voluntary and that I may withdraw at any time without consequence.
- ☐ I consent to my anonymised answers being used in this research analysis.
- ☐ *(Optional)* I consent to taking part in a follow-up interview and to it being audio-recorded.

☐ I agree to the above and wish to proceed to the questionnaire. | Researcher contact: _____

8. Data Analysis Plan & Results

The table below is the analysis plan. Statistics and *p* values are left blank deliberately and will be entered once the survey has been administered and the data cleaned.

Research question	Test	Rationale	Statistic	p	Verdict
Is adoption independent of EAP level?	Chi-square test of independence	Categorical × categorical, 4 × 2 contingency table; df = (4−1)(2−1) = 3	$\chi^2(3)$ = _____ Cramér's V = _____	_____	_____
Does weekly use differ across the four levels?	One-way ANOVA	One independent variable, four conditions, between-group design	F(3, _____) = _____ η^2 = _____	_____	_____
Is use frequency related to self-rated confidence?	Pearson correlation	Two scale variables, linear association	r = _____ r^2 = _____	_____	_____

Robustness check on the ordinal confidence item	Wilcoxon signed-rank	Ordinal measurement, normality assumption failed	z = _____	_____	_____

8.1 Descriptive results to be reported

Indicator	Result
Ever used Smart Partner	_____ % (_____ of _____ respondents)
Sessions per week among users	M = _____, SD = _____
Minutes per session	≈ _____ minutes
Used more in the week before an exam	_____ %
Most-used feature	_____
Least-used feature	_____
Primary purpose: passing the EAP exam	_____ %
Purpose: building study habits	_____ %

9. Reflection

9.1 Methodological limitations

> The sample is a convenience sample and is concentrated in the two core streams, B and C, which is where most students actually sit. The foundation and advanced streams therefore contribute far fewer respondents, and the resulting unbalanced cell sizes affect the robustness of the ANOVA — which is why non-parametric tests were also run on the ordinal items. The questionnaire is self-report, so frequency and duration are subject to recall bias; future work should cross-validate against server-side usage logs and interview High Level students to find out why they use the agent least.

9.2 What went wrong in practice

1. **Uneven level sizes** — convenience sampling left groups A and High Level thin, so the planned equal-cell comparison was impossible; we switched to reporting effect sizes and adding non-parametric tests.

2. **Ambiguous wording** — in the pilot, several students read "use frequency" as app opens rather than complete sessions; the item was redefined as "sessions" with a clarifying note.
3. **Self-report bias** — some students read "the week before the exam" as "the whole exam month", inflating duration estimates; trends are therefore reported rather than exact values.
4. **No causal claim** — a cross-sectional design cannot tell whether using the agent raised confidence or more confident students were keener to use it, so the correlation is flagged explicitly as non-causal.

9.3 What we would change next time

1. Switch to **stratified sampling** with quotas across the four EAP levels, at least 50 respondents per cell.
2. Wire in **server-side usage logs** to cross-validate self-reported frequency and duration.
3. Add a **longitudinal design**: measure confidence at the start and end of term, replacing cross-sectional correlation with a paired test.
4. Run **targeted interviews** with High Level students to explain their lower adoption.

10. Research Ethics

All participants saw the information sheet and ticked the consent form before answering. Data were anonymised, containing no names, student IDs or other identifiers, and participants could withdraw at any time without affecting their grades. Results are reported in aggregate only and are not traceable to any individual. Interview recordings, where made, were used solely for transcription and were deleted afterwards.